

Teacher:	Ray Joshua Balingkit	**Subject:**	Design & Technology
Week:	14 (Term 1)	**Year:**	10–11
Link:		**Cover:**	

LESSON OVERVIEW

What should students know/do by the end of this week's lessons?

Students will consolidate the taught design and technology concepts through an independent mock and a structured review. They will identify misconceptions, explain corrections and apply the learning to a new question or design decision. The two lessons will make individual strengths and remaining preparation needs clear before the internal school assessment.

End-of-Week Assessment/Check: How will you know they got it?

Compare original mock responses, annotated corrections and new independent work. Check whether students can explain the cause of an error and transfer the corrected reasoning to another case. Record one focused preparation need for each student.

LESSON SLOTS - TUESDAY AND WEDNESDAY

Lesson	Date	Learning Objective	The Lesson Flow	Differentiation
1	24 November 2026	By the end of this lesson, students will be able to complete a mock assessment and identify gaps in the taught design skills.	**H:** Display a short product recommendation with no evidence. Ask what a reader would need to judge it. Explain that the mock will identify which concepts and explanations require further practice. **ID:** Outline the mock coverage: user needs, product analysis, emerging technologies, lifecycle choices, energy, materials and mechanical devices, electronic systems, programmable control and measurable specifications. Explain the criteria for accurate terminology, labelled diagrams, evidence and justified decisions. Keep the original independent response separate from later supported corrections so progress remains visible. **WD:** Read the task instructions and evidence requirements together without solving the mock questions. Clarify how	**Support:** Provide readable diagrams, clear instructions and agreed classroom access support. During subsequent feedback, target the specific learning need while preserving the original independent work. **Extension:** Students identify an alternative solution and explain when it would be preferable. Ask them to test whether their original justification still holds if a key user requirement changes.

			diagrams, a short product case and a planning task should be completed and submitted. **YD:** Students complete the mock independently, including concept questions, a labelled system or mechanism diagram and a justified design decision. They mark uncertain responses and check their work against the instructions. Collect the original work and identify whether each difficulty concerns knowledge, reasoning, diagram accuracy or meeting the brief. **C:** Ask each student to identify one secure skill and one uncertain answer. Use the mock evidence to select anonymous examples and group the learning needs for the next lesson.	
2	25 Novembe r 2026	By the end of this lesson, students will be able to use mock feedback to correct a design response and apply the learning to a new task.	**H:** Show an anonymous mock error such as a reversed gear direction, a vague specification or an unsupported material choice. Ask how the error could be identified rather than simply supplying the correct answer. **ID:** Review the misconceptions shown in the mock. Distinguish a missing fact from an incorrect rule or weak justification. Explain that a useful correction states what changed and why, then checks understanding through a different example. **WD:** Model one correction on a parallel question. Trace the mechanism or control rule, link a material property to the brief or replace a vague requirement with a measurable test. Show the evidence that makes	**Support:** Group students by the misconception shown in the mock. Provide one worked example, vocabulary guide or structured diagram, then remove the scaffold for the new independent question. **Extension:** Students compare two plausible corrections, justify which better meets the brief and identify a limitation or additional test that would strengthen their answer.

			the corrected reasoning stronger. **YD:** Students annotate their original mock, explain significant corrections and complete a new independent question on the same skill. Partners check explanations against the criteria, then each student records a specific action for assessment preparation. **C:** Compare original work, corrections and the new independent attempt. Ask students to explain the reasoning they can now apply and record misconceptions that still require targeted teaching.	
3				Support: Extension:
4				Support: Extension:

MATERIALS NEEDED FOR THE WEEK

- Pearson Edexcel GCSE (9–1) Design and Technology revision notes: selected taught Core content sections
- Teacher computer, display screen and whiteboard markers
- Prepared product scenarios, diagrams and user briefs aligned to the lesson objectives
- Prepared mock tasks, model responses and feedback criteria
- Paper, pencils, rulers, card templates and approved classroom stationery
- Teacher-prepared samples or component cards; pre-cut model parts where required
- Lesson success criteria, observation checklist and labelled evidence folders; confirm Loy Krathong arrangements on Tuesday 24 November and reschedule any displaced task



END-OF-WEEK REFLECTION

- What worked well?

- What flopped or confused students?

- What needs to be re-taught or adjusted next week?